

Technical Datasheet RS485 Display

RS485 Display - Module: -

This digital LED display is designed for precise industrial data visualization. With RS485 communication and Modbus RTU support, it can interface directly with PLCs and automation systems. The display shows real-time process parameters and allows configuration of settings such as baud rate, parity, and slave ID. Its durable design ensures reliable performance in demanding industrial conditions.



Specifications

General –

I/O Connectors

2 Pin pitch pluggable screw terminals.

Dimensions

7.6 x 3.92 x 2.15 cm

Power

Input Power – 12 – 24 VDC

Typical – 12V DC @ 80mA

Operating Temperature

0 – 60° C (32 ~ 140°F)

Storage Temperature

-20 - 70° C (-4 ~ 158°F)

Storage Humidity

5 ~ 95 % RH, non – Condensing

Certifications



Display Specification:

Type	7 Segment/ 14-Segment.
Digit Count	4 Digit.
Digit Height	0.56 inch / 0.14mm.
Color	Red.
Update Rate	1Hz.
Enclosure Size	70mm x 40mm x 20mm (direct mountable in slot)

Additional Features: -

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

750V isolation.

Configuration Settings: -

Communication Speed	9600 – 19200 Kbps (Soft Serial)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	1-247
Function Code	0x03 Read Holding Registers

Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

RS485 Display Register:-

- **First Register – Value to Display**
- Purpose: Write the **actual number** you want to show on the display.
- Type: Usually a **single holding register (16-bit)**.
- Example:
- If you want to display 1234, write 1234 to this register.
- Note: The display will update immediately when the value is written.

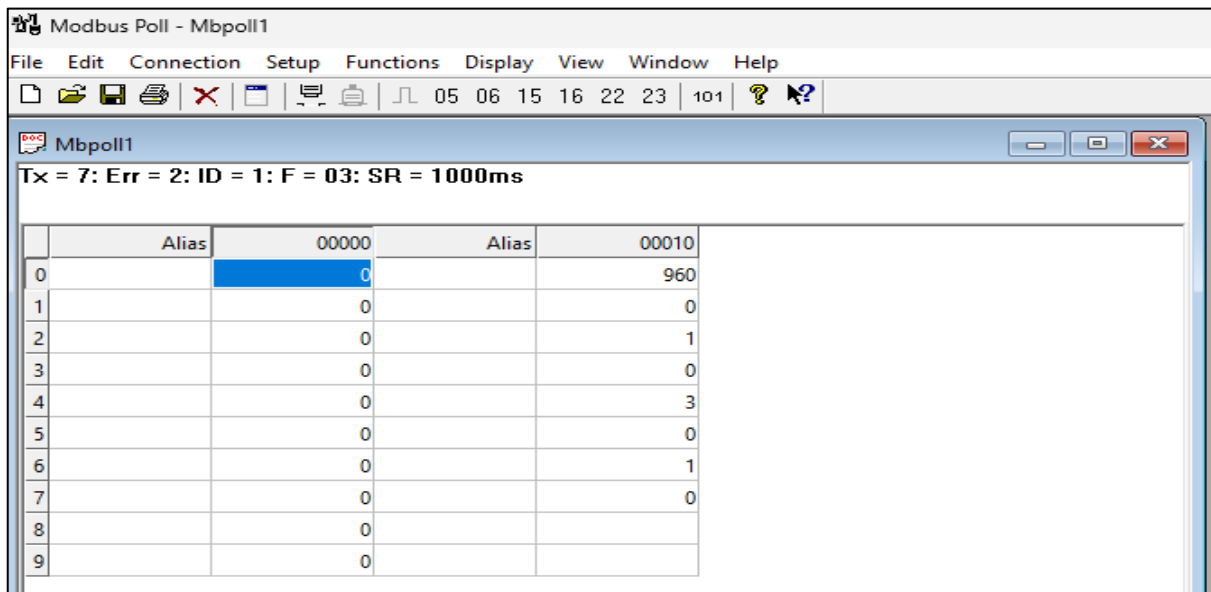
- **Second Register – Decimal Point Selection**
- Purpose: Control **where the decimal point appears** on the display.
- Type: Usually another **holding register (16-bit)**.
- Example:
 - 0 → no decimal
 - 1 → 1 decimal place (e.g., 123.4)
 - 2 → 2 decimal places (e.g., 12.34)
 - 3 → 3 decimal places (e.g., 1.234)
- Note: Some displays use **bitmasking**, where each bit represents a DP position. Always check the datasheet.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	Display Number	40001	0X03	RS485	16 Bit Unsigned int
1	Decimal Point	40002	0X03	RS485	16 Bit Unsigned int
2		40003	0X03	RS485	16 Bit Unsigned int
3		40004	0X03	RS485	16 Bit Unsigned int
4		40005	0x03	RS485	16 Bit Unsigned int
5		40006	0x03	RS485	16 Bit Unsigned int
6		40007	0X03	RS485	16 Bit Unsigned int
7		40008	0X03	RS485	16 Bit Unsigned int
8		40009	0x03	RS485	16 Bit Unsigned int
9		40010	0x03	RS485	16 Bit Unsigned int
10	Display Baud Rate (Default:960)	40011	0X03	RS485	16 Bit Unsigned int
11	Enter New Baud Rate	40012	0X03	RS485	16 Bit Unsigned int
12	Display Slave ID (Default:1)	40013	0X03	RS485	16 Bit Unsigned int
13	Enter New Slave ID	40014	0x03	RS485	16 Bit Unsigned int
14	Display Parity (Default: None-3And for Odd-1,Even-2)	40015	0x03	RS485	16 Bit Unsigned int
15	Enter New Parity	40016	0x03	RS485	16 Bit Unsigned int
16	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16 Bit Unsigned int
17	Enter New Start/ Stop Bit	40018	0x03	RS485	16 Bit Unsigned int

Note:

1. In register 40015- you will able to see the current parity and you can change it in register 40016
2. In register 40017- you will able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.

BAUD RATE and SLAVE ID DESCRIPTION:



Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 22 23 101 ? ?

Mbpoll1

Tx = 7: Err = 2: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.

To change the baud rate and slave id you have to enter a value on the 40012 resistor for the baud rate and 40014 for the slave id as shown in the image below.

Write Single Register

Slave ID: 1

Send

Address: 12

Cancel

Value: 11520

Result

N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register
☐ 16: Write multiple registers

Write Single Register

Slave ID: 1

Send

Address: 14

Cancel

Value: 25

Result

N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register
☐ 16: Write multiple registers

Mbpoll1

Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it like this 960/1920/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate(115200) and 25 at 40013 slave id.
4. You can Change Parity Bit , stop bit Of the device with same way.

Mbpoll1

Tx = 12: Err = 5: ID = 25: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		11520
1		0		0
2		0		25
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

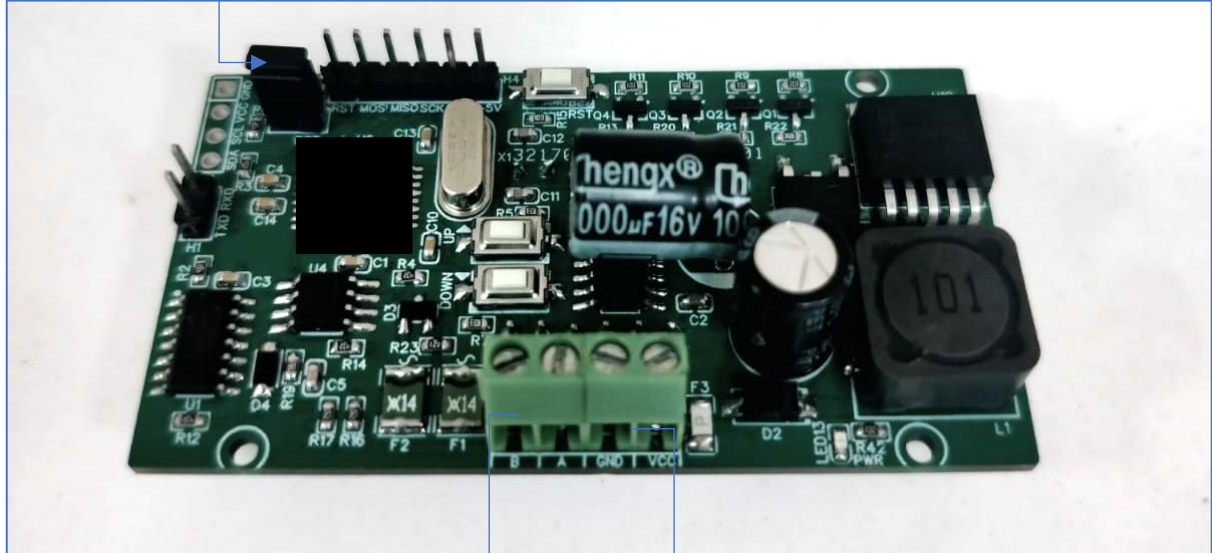
Resetting Modbus Setting

- To restore the device to its default Modbus settings:
 - Pull out the jumper** as shown in the image.
 - Press the reset button** while the jumper is removed.
 - Release the reset button.
 - Re-insert the jumper (if required).
- Note:** This will restore all Modbus settings (ID, baud rate, parity, etc.) to factory defaults. Any custom configuration will be lost.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

Reset
Jumper



Modbus
RS 485

Input
12-24
VDC

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